Arshnoor Singh Sachdeva

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SUMMARY

Mechanical Design Engineer with end-to-end ownership of precision mechanical systems for scientific instruments. Experience designing, prototyping, testing, analyzing and integrating vacuum-compatible, thermally controlled, and electromechanical subsystems in fast-paced startup and R&D environments.

TECHNICAL SKILLS

CAD: Autodesk Fusion 360, Solidworks, GD&T.
FEA: Ansys, Ansa, Abaqus, Meta, LS-Dyna, nCode DesignLife.
Programming: Python, C++, Arduino IDE.
Tools: Matlab, Simulink, KiCAD, JIRA, Confluence

WORK EXPERIENCE

ZoNexus LLC, *Product Development Engineer, Richmond, CA* **November 2023 – Present**

- Owned end-to-end development of mechanical subsystems from concept and detailed design through prototyping, testing, and system integration in a fast-paced startup environment.
- Contributed to successful $1.1M Department of Energy SBIR Phase II award by performing thermal simulations and redesigning internal components, reducing system thermal drift by 22%.
- Designed CNC-machined and fabricated assemblies using DFM/DFA principles to balance tolerance stack-up, robustness, and assembly efficiency.
- Integrated electromechanical hardware–including PCBs, sensors, motors and controllers–into complete systems, supporting bring-up, debugging, and failure resolution.
- Generated detailed 2D drawings with GD&T to ensure repeatable builds and supplier manufacturability.
- Developed and documented assembly workflows and test procedures to enable consistent prototype builds and transition R&D hardware toward production readiness.
- Collaborated with suppliers and machining team on component sourcing, prototype builds, and iterative design improvements based on test results.
- Performed hands-on troubleshooting and repair of laboratory equipment, diagnosing mechanical, vacuum, and electromechanical faults to restore system functionality and maintain operational uptime.

Tesla Inc, *Analysis Engineering (CAE) Intern, Fremont, CA* **May 2022 – August 2022**

- Built and validated finite element models to evaluate durability, fatigue life, and vibration performance of chassis subsystems, including brake pedals, suspension components, and motor mounts.
- Identified bulk deflection source in brake pedal assembly as the surrounding dash panel and proposed targeted reinforcements, achieving a 30% stiffness improvement and more robust pedal feel.
- Analyzed load cases and failure modes to identify fatigue and NVH risks, and proporsed geometry and material changes to improve structural life and vibrational response.
- Partnered with design engineers to translate CAE results into production-intent design updates, supporting data-driven decisions under program timing and manufacturing constraints.

EDUCATION

Arizona State University, Tempe, Arizona, USA **August 2021 - May 2023**
Master of Science in Robotics (Mechanical Engineering) **GPA: 4.00/4.00**
Birla Institute of Technology & Science Pilani, India **August 2017 - May 2021**
Bachelor of Engineering in Mechanical Engineering **GPA:3.7/4.0**

PROJECTS

Vacuum compatible water-cooled Peltier module **ZoNexus LLC**

- Designed and built a vacuum-compatible water-cooled Peltier module for temperature-controlled instrumentation.
- Developed empirical bench test setup to characterize thermal behavior and drive design iteration.
- Engineered modular precision mounting interfaces enabling multi-platform integration with minimal drift.
- Identified interface thermal resistance as performance bottleneck and improved stability through interface material and mounting redesign.
- Integrated system into vacuum chamber and achieved stable sub -60°C.